Literature Review for Master Thesis

1 Introduction for Thesis

- **General Introduction:** what are avalanches, how do they form?

  Avalanches are rapid masses of snow, which move down mountain slopes due to gravity (Ancey, 2001). Snow avalanches can either be loose snow, which starts from a point in a cohesionless surface, or snow slab avalanches (Schweizer et al., 2003). In this Master Thesis, the focus will be on dry snow slab avalanches. Snow slab avalanches consist of a cohesive slab of snow (Schweizer et al., 2003), which are released because of loading at the surface, which triggers crack propagation in a weak layer (Gaume et al., 2017). Weak layers are characzetized by the grain shape of snow, which leads to lower cohesion of those layers (Föhn et al., 1998).

- **Relevance:** Why model avalanches?

  Snow avalanches are a major natural hazard in mountainous region, not only do they endanger human life but also infrastructure (Schweizer et al., 2003). The Swiss institute for snow and avalanche research has reported an average of 22 fatalities caused by avalanches every year with the number of non-fatal accidents being tenfolds higher (SLF, 2025). This data shows that it is crucial to understand how avalanches form and propagate in order to protect people and infrastructure better.

- **Motivation:** why avalanches & DAS, how is this different from papers discussed in introduction

  In this master thesis, the aim is to detect avalanche slab breakoff and crack propagation using distributed acoustic sensing (DAS).

  In DAS, an optical signal is sent, the backscattering of this signal is measured (Shatalin et al., 1998). In distributed acoustic sensing, all points along the fibre are measured at the same time (Parker et al., 2014). The dynamic strain is estimated by measuring the phase variations of the backscattered light (Hartog, 2017). From the strain, the deformation can be obtained (Trabattoni et al., 2023). In other words, this means that a perturbation close to the DAS cable such as the breakoff of a slab avalanche causes the phase in the backscattered light to change, which is translated to deformation of the cable.

  As mentioned, avalanches are a major threat to human life in mountainous regions, this is why detecting the breakoff of the snow slab is important for hazard mitigation. Detecting the breakoff of the slab means that there can be warning at lower parts of the mountain that there is a possible avalanche. Furthermore, detecting slab breakoff with DAS is also important to further understand crack dynamics in snow as well as the factors influencing crack propagation.

  A pilot early-waring system using distributed acoustic sensing has been implemented in Norway, where avalanches are detected by their seismic signature (Turquet et al., 2024), meaning when the avalanche is already moving downhill. The detectability of avalanches by DAS has also been studied in the Swiss alps by Paitz et al., 2023 and Edme et al., 2023, but the detectability of the slab breakoff itself has not been studied.

- **Modeling methods:** Introduce snowpack/crocus, why are they relevant, why will they not be used, introduce MPM, DEM, advantages & disadvantages, introduce salvus and why is this needed

- **Previous models:** Introduce previous models, methods used, limitations & further developmemts

  A multitude of models are used to study different processes in snow. The SNOWPACK and CROCUS models used in the Swiss and French alps model the structure and evolution of snow (Bartelt & Lehning, 2002; Brun et al., 1992). These models were developed to understand how snow evolves under different conditions, for example a changing temperature gradient and water content and what consequences this has for the formation of weak layers where failure likely occurs and avalanches are triggered (Bartelt & Lehning, 2002; Brun et al., 1992). SNOWPACK and CROCUS are comprehensive models that take into account multiple governing equations and processes and what consequences they have for the structure of snow. While it is important to understand the influences on for example temperature gradient on the snow structure and weak layer formation and stability as a consequence, this is not the focus of the thesis. This is because these models are too complex and comprehensive for the purpose of trying to detect slab breakoff in DAS data as they model the entire snow structure. Using SNOWPACK or CROCUS to investigate how different snowpack compositions and factors influence the DAS signal is beyond the scope of this thesis.

To model crack propagation in snow, multiple numerical modeling methods have been used. Cundall and Strack, 1979 developed the distinct element method (DEM) for granular media. This model discretizes the domain as spheres and disks, which interact with each other based on Newton's second law (Cundall & Strack, 1979). The interactions of particles are modeled by the governing equations and the resulting forces and displacements (Cundall & Strack, 1979). This method has been used by **bbobillierMicromechanicalModelingSnow2020**; Bobillier et al., 2024; Mulak and Gaume, 2019 and many others to model crack propagation and its influencing factors. The main limitation of DEM is that the particle contacts and the corresponding equations need to be recalculated every time a grain contact is made or broken (Cundall & Strack, 1979). This means that for large, slope-scale simulations, this method becomes increasingly computationally expensive. In addition to the high computation cost, DEM also cannot account for anticrack propagation, the closure of cracks in this propagation mode is accounted for by manually removing elements from the mesh, which is not very efficient (Gaume et al., 2019).

This problem is solved by using the material point method (MPM). In this method, the material is discretized in a set of Lagrangian points, which are used to track mass, deformation and momentum (Bardenhagen et al., 2000). Because the points are not fixed, an Eularian background mesh is needed to calculate spatial derivatives (Bardenhagen et al., 2000). This method is not only more computationally efficient on the slope-scale, it can also account for anticrack behavior and strain softening of snow (Gaume et al., 2019). This is why the material point method will also be used in the thesis to investigate slab breakoff in DAS data on a large scale. not sure if this next part is necessaryThe material point method has been used by Bergfeld et al., 2025, who modeled slope-scale experiments to understand the transition of crack propagation modes, Trottet et al., 2022 who modeled propagation saw tests to observe transitions in crack propagation speeds, as well as Gaume et al., 2019 and Gaume et al., 2018.

does this need to be more detailed? like explain what spectral elements is as well and more how salvus works? In order to understand how crack propagation and slab breakoff would look like in DAS data, numerical modeling is done. This is done in SALVUS, which is a package which utilizes spectral elements to model seismic wavefields (Afanasiev et al., 2017).

- **Crack Propagation:** intro to modes of crack propagation, where and how cracks propagate, influencing factors like snow properties, slope angle etc, cracking modes transitions & influences on transitions

  While it is important to understand the different modeling methods and their limitations, it is also important to understand the different types of crack propagation and what factors influence them. Understanding how fast a crack can propagate and which factors influence this is important for warning, as it relates the time of the initial crack to the time where the slab breaks off depending on snow and terrain parameters.

  Two main types of crack propagation are believed to be important for weak layer failure: anticrack propagation and shear crack propagation (Bergfeld et al., 2025). There are three ways which cracks can open up: mode I, where space is opened up between the crack faces under tensile loading, mode II and mode III which represent paralell shearing and diagonal shearing (Heierli et al., 2003). In anticrack propagation, the weak layer consolidates under loading (Knight & Knight, 1972), while cracks propagate in a combination of modes I-III (Heierli et al., 2003). Gaume et al., 2018 modeled anticrack propagation induced by a propagation saw test (PST) using MPM, which showed that the scale of the simulations influences the propagation direction of the cracks. For smaller scale simulations, cracks propagate from top to bottom of a slope, whereas for slope-scale simulations, cracks propagate bottom to top, which means that remote triggering of avalanches is possible (Gaume et al., 2018). This is an important consideration for the thesis because it influences where the slab breakoff would be observed in terms of DAS cable location. Other models also show that the crack propagation speed of anticracks are influenced by slope angle and curvature: concave slope areas have higher propagation speeds than convex areas of the slope (Gaume et al., 2019). Furthermore, there is a directionality in propagation speed, simulations by Gaume et al., 2019 show that cracks propagate twice as fast upslope than in the downslope direction, which is especially important when thinking about remote triggering. The slab breaks off when the slope angle exceeds the friction angle of the material (Gaume et al., 2019). Slope angle also influences which mode of crack propagation is more dominant (Rosendahl & Weißgraeber, 2020). In steeper terrains, shear modes (II and III) dominate, whereas in flatter terrain, mode I dominates the crack propagation (Rosendahl & Weißgraeber, 2020). The critical force (skier force) needed for weak layer failure is also dependent on slope angle, it decreases with increasing slope angle (Rosendahl & Weißgraeber, 2020). Snow parameters also have an influence on this critical force, as shown by simulations by Rosendahl and Weißgraeber, 2020: the thicker the slab or the higher the slab tensile strength, the better the force by the skier can be distributed on the weak layer and there is more force needed for failure.

Multiple experiments and numerical models have shown that cracks can propagate faster than the elastic wave speed in snow, so called supershear crack propagation (Bergfeld et al., 2025; Bobillier et al., 2024; Trottet et al., 2022). Understanding what causes the transition from sub-Rayleigh to supershear crack propagation is of high importance of the thesis, because it can give an estimate on whether supershear crack propagation will occur under certain conditions, which leads to faster propagation of cracks and less time between loading and slab breakoff. In MPM models of PST, Trottet et al., 2022 observe that the occurence of supershear cracks is dependent on slope angle: in steeper terrains, supershear propagation was observed. This can be explained by slab bending, because in steep terrains, bending of the slab is not limited to the collapse depth of the weak layer, slab tension can increase more, which leads to faster crack propagation (Trottet et al., 2022). These observations were also made by Bobillier et al., 2024 as well as Bergfeld et al., 2025. The release area of an avalanche is also tied to propagation speed, at supershear propagation speed, there is less crack arrest and therefore larger release areas (Bergfeld et al., 2025; Trottet et al., 2022). This is crucial because larger release areas also mean a larger snow volume, which has more potential for damage.

- **Considerations relevant for thesis**: how does scale influence models, what factors influence DAS modelling, open questions

2 Synthesis

The papers discussed can be divided into two methods utilized to model snow processes. Understanding how snow can be modeled accurately and efficiently is very relevant for the thesis, because a large part of the thesis will be modeling DAS data of avalanches in snow. This is why it is important to understand the different methods used to model snow and their advantages and disadvantages as well as the physical processes underlying these models. A majority of the papers discussed below utilize the discrete element method described by Cundall and Strack, 1979. In this method, the interaction between individual grains is modeled by calculating the forces and the resulting displacements on the grains (Cundall & Strack, 1979). This method is both used to describe snow failure as well as crack propagation in snow, for example Gaume et al., 2015 model crack propagation in a weak snow layer using discrete elements. This modeling method is accurate in modeling snow processes, but as described by Bobillier et al., 2020, it models snow on the grain scale, which is computationally expensive when modeling snow on the slope scale. Bobillier et al., 2020 developed a model on the particle scale, which is still accurate at a higher computational efficiency. Another method used by some papers is the material point method. This method discretizes the material into a set of material points, which are used to calculate the deformation and stress of the material (Gaume et al., 2018), the time dependent equations can only be solved by using an Eulerian background mesh. This method is used by multiple papers, to model crack propagation, for example Gaume et al., 2018 use this method to model crack propagation in snow. The MPM method is advantageous over the discrete element method because it is more computationally efficient, which means that it can be used to model larger scale processes such as slope scale avalanches, whereas the discrete element method is more accurate but computationally expensive, which means that it is more suitable for modeling smaller scale processes such as crack propagation in a weak layer. In MPM, contacts are modeled in the background grid, whereas in DEM they are modeled individually and every time a contact is made or broken, the stiffnesses need to be recalculated. The MPM method also allows for modeling of strain softening, more easily, this is important when modeling snow processes. Modeling strain softening using discrete element methods is possible, but more difficult because elements need to be removed manually to account for the cracks closing, whereas in the MPM method, strain softening can be modeled by a yield surface with varying pressure, which allows for strain softening even at macroscopic uniaxial compression (Gaume et al., 2018). Snow can also be modeled by physical models for snowpacks, such as SNOWPACK described by Bartelt and Lehning, 2002 or CROCUS described by Brun et al.,

1992. These models implement processes such as snow metamorphism and settling of snow, which are important to understand where weak layers form. Both models developed by Brun et al., 1992 and Bartelt and Lehning, 2002 account for changes in microstructure of snow and the effects of snow layering and temperature gradient changes. Understanding what effect these processes have on snow structure is important to understand where avalanches will likely form, but it is not necessary to model these processes in the thesis, because the focus is not on snowpack evolution. Mulak and Gaume, 2019 show that sintering, which means the formation of new bonds between snow grains, is an important process to consider when modeling snow failure, because it can increase the strength of the material and delay failure. This is important to consider for the thesis because it can mean that there is a large time between loading and failure, which can influence whether cracks are observed in DAS data and how long it takes between initial cracking and avalanche release. However, Mulak and Gaume, 2019 also show that if there is rapid loading, sintering is too slow to prevent failure, which means that in cases of rapid loading, sintering does not need to be considered. This is relevant for the thesis because in cases of rapid loading, such as a skier triggering an avalanche, sintering does not need to be considered, whereas in cases of slow loading, such as a large accumulation of snow, sintering needs to be considered. Such processes are neglected in the model developed by Rosendahl and Weißgraeber, 2020 as well as the model by Gaume et al., 2019, who develop models only for the fast-loading case. Understanding the effect of sintering is important to represent snow processes accurately in the thesis: does the loading occur over large times (by snow itself) or is it a rapid loading event (skier or explosion).

- Could different models be made depending on loading rate to compare DAS data?
- Could DAs data look different for rapid and slow loading?

Multiple papers show that after loading, there are different stages of deformation of snow. Gaume et al., 2018 showed that snow responds to loading by an elastic regime, a drop in pressure and shear stress, volumetric collapse of the weak layer and lastly dense packing. do these stages show up individually in DAS data or are they one event?. Bergfeld et al., 2025 also discuss different deformations, but mainly related to the location in the snowpack. At the lower areas, deformation is because of slab bending caused by weak layer collapse, whereas higher in the snowpack, deformation is because of tension of the slab. Several papers also highlight that not all cracks trigger release: a crack must reach a critical length before a slab can detach (Gaume et al., 2017), and the required length is sensitive to slab stiffness, slab thickness, weak-layer thickness, slab density, and slope angle. The weaker the snowpack, the lesser the crack length needed for failure. Related to this, crack growth can occur in different modes: mode I (opening) versus mode II (shear), and mixed-mode loading changes the critical stress for propagation (Rosendahl & Weißgraeber, 2020). These concepts are important for DAS interpretation because only cracks that approach (or exceed) their critical length are likely to produce a large enough deformation signal and to lead to avalanche release, whether this happens is related to snow parameters.

All the papers discussed below show that avalanches are triggered by initial loading, which causes failure in a weak layer, which then propagates through the snowpack. Understanding how the failure or cracks propagate through the weak layer and how this is modeled is important to model DAS data because the type of avalanche, the time between loading and release and whether there is an avalanche at all depends on the type of failure and crack propagation. Multiple papers discuss the type of crack propagation in snow, for example anticrack propagation, as well as the different processes influencing crack propagation. Gaume et al., 2015 show that the type of crack propagation in snow is influenced by the slope angle, the elastic modulus of the slab and the elastic modulus of the weak layer. If young's modulus of the slab is high, the density of the snow is high or the weak layer thickness is high, cracks propagate less far because failure occurs faster, which means that there is less time for cracks to propagate. If the tensile strength of the slab is high, crack propagation speed is higher, because the slab can sustain more stress before failure occurs, which means that there is more time for cracks to propagate. If the weak layer is very weak, crack propagation speed is also very low, because failure occurs before cracks have propagated. Direct failure also occurs when the slab is very soft (Bobillier et al., 2024). Furthermore, Gaume et al., 2017 show that the type of crack propagation is influenced by the slope angle, on flatter terrain, the bending of the slab is limited by the collapse depth, which leads to a constant speed of crack propagation, whereas in inclined terrains, slab tension can increase, which also leads to crack propagation speeds to increase (Trottet et al., 2022). Bobillier et al., 2024 also show that slope angle is important for the crack propagation speed. At high slope angles, Bobillier et al., 2024 observe crack propagation speeds higher than the shear velocity in the material (supershear), whereas at lower slope angles, the propagation speed of cracks is lower (sub-Rayleigh). As described by Trottet et al., 2022, the faster cracks propagate, the faster the slab breakoff will occur, which means there is less time between the initial cracking and the release of the avalanche. Bergfeld et al., 2025 make the link between shear crack propagation and anticrack propagation. In their simulations and lab experiments, Bergfeld et al., 2025 show that early stages of crack propagation are anticrack-dominated, while at later stages, shear cracks occur. Again, if the system reaches supershear propagation is dependent on multiple factors, as previously mentioned. The size of the release area observed by Bergfeld et al., 2025 is dependent on crack arrest, which is dependent on the propagation regime. Supershear speeds only show crack arrests because of topography, which leads to larger release areas (Bergfeld et al., 2025). The relationship between crack propagation regime and release area is important because it can help making more accurate risk assessments based on DAS data: if the crack propagation is supershear, there is a higher risk of a large release area, which means that the avalanche can have higher volume. Multiple papers also mentioned that the propagation direction crack propagation is dependent on the scale of the simulation. Understanding in which direction cracks propagate is not only important to understand where the slab will break off, it is also important to understand where in relation to the DAS cable the breakoff event can be observed. Gaume et al., 2019 show that in small scale simulations, cracks propagate upslope, whereas in larger scale simulations, cracks propagate downslope. This is explained by Trottet et al., 2022 by the fact that in small scale simulations, the bending of the slab is limited by the collapse depth, which leads to a constant speed of crack propagation, whereas in inclined terrains, slab tension can increase, which also leads to crack propagation speeds to increase (Trottet et al., 2022).

3 Avalanche & Snow Modeling

3.1 Micromechanical modeling of snow failure (Bobillier et al., 2020)

Main Question of article: Can crack and mechanical behavior be represented in a computationally efficient model on the slope scale?
Why this is relevant for the thesis: Understanding how to model snow and snow mechanical behavior is important to model DAS data in snow accurately.
Models previous to Bobillier et al., 2020 represent crack behavior accurately, but on the grain scale, which is very computationally inefficient when simulating slope-scale processes. Bobillier et al., 2020 used a discrete element model on the particle scale, which acts as a mesoscale between the grain level and the snow sample scale. The model developed agrees with data from cold-lab experiments. The model used is a three layer model with a rigid basal layer, a weak layer and a slab layer (Bobillier et al., 2020). Under tensional regimes, Bobillier et al., 2020 observed elastic-brittle failure in snow, whereas the weak layer, under mixed loading, showed four distinct deformation stages. The stages can be described as elastic deformation, strain softening which happens at the same time as brittle crushing of particles, followed by shear displacement (Bobillier et al., 2020).

3.2 Numerical investigation of the mixed-mode failure of snow (Mulak & Gaume, 2019)

Main Question of article: What influence do different loading angles and rates have on failure criteria of snow layers and what effect does sintering have on this?
Why this is relevant for the thesis: Understanding the effect of loading rates, loading angles and failure modes is very important to understand effects seen in DAS data: this influences the type of avalanche, the time between loading and release and whether there is an avalanche at all
Snow material failure resulting in avalanches occurs on the individual bond scale, simultaneous shear and compressive loading further increases the complexity (Mulak & Gaume, 2019). Understanding which loading angles and rates lead to which types of failure is of importance in understanding DAS data of avalanche release, because as Mulak and Gaume, 2019 show, these processes are quite complex. Sintering is an important process in understanding material failure, because broken bonds can be reformed, which can even increase the strength of material. Mulak and Gaume, 2019 state that if there is high stress, compression of the material increases, which leads to an increase in strength of the material and the failure is delayed. This has relevance for the thesis because there can be a large amount of time between the loading and the actual material failure. If there is rapid loading on the other hand, sintering is too slow and there is still failure (Mulak & Gaume, 2019). Sintering in the avalanche case would occur if there is a large accumulation of snow and a slow increase in compressive load (Mulak & Gaume, 2019).
Could the DAS data look different for rapid loading and slow loading because the stresses are different?
Mulak and Gaume, 2019 implemented a discrete element model where different failure conditions are

investigated at loading angles between 0° and 180°. Failure is defined by an increase in average velocity of the slab (meaning the slab would slide) or an increase in maximum value of the stress (Mulak & Gaume, 2019). Mulak and Gaume, 2019 implemented sintering in their model by the creation of new bonds in the discrete element structure. For simulations with no sintering, at low loading angles, there is a decrease in cohesive bonds, which further increases with higher loading angles (Mulak & Gaume, 2019). This means that at low loading angles, the number of broken bonds required for material failure is much larger, as stated by Mulak and Gaume, 2019. At higher loading angles, the total strength of the weak layer decreases, which means that the shear stress is dependent on loading angle (Mulak & Gaume, 2019). Where sintering was implemented by Mulak and Gaume, 2019, the results are almost the same as in the case without sintering. The number of broken bonds needed for failure is larger however, because new bonds can form, and the strength decreases more if the loading angle increases (Mulak & Gaume, 2019). Mulak and Gaume, 2019 show that sintering is most important in compression-dominant modes at low loading angles. If sintering times are low, the failure of material occurs faster than the formation of new bonds, which means that the results without sintering are the same as the ones without sintering (Mulak & Gaume, 2019). All simulation data show good agreement with lab data. Overall, Mulak and Gaume, 2019 also show that shear strength of a material is more crucial in material failure compared to compressive strength, because shear strength is often lower. This means that the shear load is more likely to cause failure as opposed to the compressive load (Mulak & Gaume, 2019). The models developed by Mulak and Gaume, 2019 are only 2D, which is a limitation because porosity is limited, this limits strain softening and possible collapses of weak layers.

3.3 A physical SNOWPACK model for the Swiss avalanche warning: Part I: numerical model (Bartelt & Lehning, 2002)

Main Question of article: How can the snowpack and its governing equations be modeled accurately but as simple as possible?
Why this is relevant for the thesis: Modeling snowpack evolution and structure is important to forecast avalanches (where are avalanches likely to form), but also to understand avalanche dynamics (in which layers do avalanches form)
Bartelt and Lehning, 2002 describe that numerical modeling of the snowpack is important for avalanche prediction. The snowpack-model developed solves the partially governing equations using finite-element methods (Bartelt & Lehning, 2002). Understanding snowpack evolution and avalanche formation is also relevant for the thesis because the crack propagation which is modelled in the thesis depends on snow properties. According to Bartelt and Lehning, 2002, most avalanches occur during or after periods of intense snowfall. Snow settling gives an insight on mechanical stability of snow, which is also important to understand crack propagation: in stable layers cracks probably do not propagate as well as in layers which already have some instabilities. The microstructure and the layering of snow also needs to be modeled to detect discontinuities or layer boundaries, which is important to assess stability of layers (Bartelt & Lehning, 2002). Bartelt and Lehning, 2002 also model the influence of solar radiation and warm winds on the energy exchange, the melting of snow and the subsequent transport of meltwater is important to understand the release of wet snow avalanches. Existing snowpack models such as CROCUS rely on Truesdell's equation for mixture and postulate conservation of ice, water, water vapor and air of the snowpack, these models are simplified (Bartelt & Lehning, 2002). This means that the existing models for example do not describe settling of the snowpack, they assume the snowpack to be dry or the absence of snow metamorphism, which does not depict the reality of the snowpack accurately (Bartelt & Lehning, 2002). An accurate and physical model of the snowpack is important for the thesis because modelled data should be comparable to field data measured by the DAS cable. This would not be the case if simplifications are made in the simulation of crack propagation in snow, the simulated data would look different from the measured data. The novel SNOWPACK model described by Bartelt and Lehning, 2002 uses three constituents (ice, water and moist air) to describe the snow. These three all have the same temperature, which is the snow bulk temperature measured in meteorological observations, assume all meltwater processes to be energy and mass conserving and assume that the weight of the snowpack is carried by a single ice lattice (Bartelt & Lehning, 2002). Bartelt and Lehning, 2002 use four governing equations to describe the snowpack: the energy transfer equation at the surface (conservation of mass of vapor and water and the bulk temperature), because all of the snow constituents have the same temperature, a momentum equation for settling of the snow because the weight of the snow is carried by a single ice lattice: The conservation of mass of ice is not considered because the numerical solution of the displacement is continuous because of the Lagrangian coordinate system. Any change in the snow

load results in a change of the stress state of the settlement equation. Snow cannot be assumed to be a plastic material because deformations caused by loading are irreversible, understanding this is important for crack formation and propagation. Bartelt and Lehning, 2002 describe a microstructure-based viscosity formulation, plasticity theory is not applicable because of time-dependent yield surfaces and a directionality of plastic flow which is defined in relation to the yield surface. The initial conditions of the snowpack such as snow height are taken from meteorological data, the energy exchange at the surface is the most important boundary condition (Bartelt & Lehning, 2002). Neumann boundary conditions are used if the temperature at the surface is close to the melting temperature, if the temperature is well below the melting temperature, Dirichlet boundary conditions are used (Bartelt & Lehning, 2002). The melting and refreezing processes in the snowpack are essential in modelling snow structure changes, which have an effect on DAS signal propagation as well as avalanche formation. Bartelt and Lehning, 2002 model these by the numerical heat transfer solution. Because the snow temperature is assumed to be constant 0°C, if the heat transfer solution is greater than 0°C, the excess energy is assumed to be used for melting, if the heat transfer is smaller than 0°C, the energy is assumed to be used for refreezing (Bartelt & Lehning, 2002). Bartelt and Lehning, 2002 used the snow structure measurements of winter 1999, which was a winter with numerous avalanches, to verify the SNOWPACK model developed. Generally, there is good agreement between the measured and modelled data, which means that the model is physical, with some exceptions (Bartelt & Lehning, 2002). If the calculated snow height is different from the measured snow height, the model corrects this by adding elements to the finite element grid in case the calculated height is too low, or removing elements if snow height is overestimated (Bartelt & Lehning, 2002). This leads to mistakes if the snowpack settles quickly: if there is quick settlement, snow height decreases quickly because of densification, the model would remove elements at the surface without accounting for densification of the snow at the bottom (Bartelt & Lehning, 2002). Snow density and snow settlement have strong effects on heat conductivity of snow, Bartelt and Lehning, 2002 mention that errors in either of these fields have strong effect on the entire simulation. The SNOWPACK model developed by Bartelt and Lehning, 2002 includes modeling of the microstructure of snow as well as density discontinuities, which represent layer boundaries, which is important to understand snow structure as well as crack propagation related to avalanches. The multicomponent description developed allows for representation of mass and energy conservation (Bartelt & Lehning, 2002), which can be used to assess snow metamorphism related to temperature and moisture, but also to determine where wet snow avalanches form due to the presence of water. Bartelt and Lehning, 2002 describe the volumetric deformation of fresh snow by large strain description, the thermal boundary conditions which need to be used can be determined by meteorological data. These data can also be used to determine if there is melting or refreezing of snow, which again is important for snow structure and stability. The laws for heat conductivity and viscosity related to microstructure show the relations between densification of snow and heat transfer (Bartelt & Lehning, 2002).

3.4 A numerical model to simulate snow-cover stratigraphy for operational avalanche forecasting (Brun et al., 1992)

Main Question of article: How do snow conditions such as temperature gradient and water content influence snow metamorphism and how can this be implemented in the CROCUS-snowpack model?
Why this is relevant for the thesis: Snow metamorphism changes grain properties, which is important for the formation of weak layers which can trigger avalanche release, but also overall stability of the snowpack. Furthermore, layers of different grain properties will depict DAS signals differently.
When a snowpack is subjected to a change, like a change in temperature or pressure conditions, the snow grains change (Brun et al., 1992). Snow metamorphism changes the grain sizes and shapes, which changes mechanical stability of the snow (Brun et al., 1992). This is relevant for avalanche modelling because certain types of snow metamorphism can weaken layers in snow, such layers can act as failure planes and make release of avalanches easier. Snow metamorphism is affected by the water content of the snow, the capillary pressure of the water between the snow grains, the radius of curvature of the grains (shape), as well as the temperature gradient (Brun et al., 1992). Brun et al., 1992 describes the main limitation of past metamorphism models as assuming the snowpack to have homogeneous shapes and sizes, which are often spherical or simplified geometric shapes, this does not represent snow accurately however, especially fresh snow is very heterogeneous. Brun et al., 1992 conducted experiments for different snow types and conditions to determine new equations to describe metamorphism. These experiments showed that the rate and type of metamorphism does not depend on crystal type but rather on the temperature gradient (Brun et al., 1992). A gradient below 5°C/m results in rounded crystals, a

gradient above 5°C/m results in faceted crystals, which transform to depth hoar, if a gradient of more than 15°C/m is applied (Brun et al., 1992). This is relevant for avalanche modelling, because it is critical to understand how snow grains evolve to assess their stability, which can be used to determine where weak layers will form. These experiments were conducted for dry and wet snow conditions. Brun et al., 1992 describe a new description of snow, which takes into account the dendricity, sphericity and size of the grains. When snow falls, new layers with different properties are added to the snowpack (Brun et al., 1992). Simulating these different layers is again important to assess the stability as well as interactions of the different layers. Brun et al., 1992 combined the observations made in the laboratory with the CROCUS snowpack model to simulate these different layers based on meteorological observations. Using MEPRA (a system for avalanche risk forecasting), the shear stress of the different layers was estimated at a 45° angle. This is important for avalanche forecasting, because the strength of different layers can be assessed and it can be determined under which stress the layers fail.

3.5 A discrete numerical model for granular assemblies (Cundall & Strack, 1979)

Main Question of article: How can the forces and the resulting displacements be modeled in a granular medium?
Why this is relevant for the thesis: Snow consists of grains, forces at the surface such as mechanical loading can cause displacement of the individual grains. If this happens at a very large scale, it is an avalanche.
The distinct element method described by Cundall and Strack, 1979 is a numerical modeling method to model the mechanical behavior of a material composed of disks & spheres. This is relevant to the thesis because snow is composed of individual snow grains, which interact with each other at their contacts, all the grains together form the snow pack. It is important to model the behavior and displacement of the grains to model avalanche dynamics because avalanches form through loading or failure in a weak layer, which then propagates through the entire snowpack (Gaume et al., 2017). Since a snowpack is a medium composed by individual snow grains, which are not entirely compacted (contain air between the grains) and interact with each other at grain contacts, the medium described by Cundall and Strack, 1979 represents snow well. It is a granular medium composed of distinct particles, which interact with each other at their contacts and form a porous medium (Cundall & Strack, 1979). The loading and unloading of such materials is complex and stresses cannot be measured easily because of the interactions of different grains (Cundall & Strack, 1979). The distinct element method can model particles of any shape (Cundall & Strack, 1979), which is important for snow modelling because the snow grains are not always spherical. Cundall and Strack, 1979 model the transient interaction of particles by calculating the contact forces and displacements of the stressed discs (particles) caused by disturbances at the boundaries. In the avalanche case discussed in the thesis, the disturbance is represented by loading of the surface, which can be caused by a skier, which causes failure in a weak layer, or cracking in the snowpack. This results in stress propagation through the snowpack, which results in displacement of the particles, which is the avalanche motion. The resultant forces on the disks are calculated only by the discs which are in direct contact (all the disks that touch one disk cause its resultant force) (Cundall & Strack, 1979). Cundall and Strack, 1979 describe a model only for dry materials, this means this is only applicable for dry snow avalanches which contain no water between the snow grains. Numerical modelling, such as described by Cundall and Strack, 1979 is advantageous over lab tests and experiments because it is less time consuming and any data can be accessed at any stage, which means that parameters such as loading, particle size, and properties such as stiffness and shape can be changed at any stage. In the avalanche case, this means that simulations can be run for different snow properties and loading scenarios, for example a simulation for older, more compacted snow can be run as well as a simulation for fresh snow. The numerical model for spheres is based on methods described by Serrano and Rodriguez-Ortiz (1973, I CANNOT FIND THIS PAPER ONLINE). The forces at the interfaces between the spheres are calculated by incremental displacements of the centers of particles, where shape changes of the individual particles are negligible (Cundall & Strack, 1979). How can the center of the grain be determined if the grain has an irregular shape? The matrix representing contact stiffness between the grains described by Cundall and Strack, 1979 needs to be recalculated whenever a new contact is made or broken, meaning when there is a crack in the snowpack or when new contacts between snow grains are made. The calculations of forces are based on Newton's 2nd law, which describes the movement of a particle as the result of the forces acting on it. The deformation is neglected because the deformation of individual particles is small compared to the deformation of the medium as a whole (Cundall & Strack, 1979). This means that the

deformation of the individual snow grains is small compared to the deformation of the entire snowpack which represents the avalanche movement. What about the fusing / compaction of grains which changes snow properties? The relative displacement of a particle is calculated by the sum of the force increments acting on it, particles can also overlap, in this case the contact forces and the interaction forces also need to be calculated (Cundall & Strack, 1979). The location of the rigid boundaries (walls) described by Cundall and Strack, 1979 needs to be specified before modelling. This could be a possible limitation for the avalanche case because there is only one rigid "wall" which is the interface of snow and the rock below it.

4 Crack Propagation

4.1 Signatures of the sub-Rayleigh to supershear fracture transition in snow avalanche experiments (Bergfeld et al., 2025)

Main Question of article: Can supershear crack propagation be observed in snow avalanche field experiments and are there different stages of crack propagation?
Why this is relevant for the thesis: Possible different stages of crack propagation speeds and their transitions are important for DAS data interpretation so that accurate risk assessments can be made.
Previous to Bergfeld et al., 2025, supershear fractures in snow have been observed in many numerical models, but never in slope experiments. Bergfeld et al., 2025 conducted controlled experiments on a snow-covered slope and did optical tracking of cracks, they also used MPM modeling to validate the slope experiments. Both the experiments and the numerical models showed that there are different stages in crack propagation (Bergfeld et al., 2025). Understanding the different stages of crack propagation speeds and what causes the transitions between them is important to make accurate risk assessments based on DAS data: do cracks propagate slowly which causes a slow breakoff of the slab or is the propagation supershear? The different stages of crack propagation observed by Bergfeld et al., 2025 correspond with different locations in the snow: at the lower end of the snowpack, the deformation was dominated by slab bending induced by weak-layer collapse, whereas at the higher end, the deformation was dominated by slab tension which was induced by the shear failure of the weak layer (Bergfeld et al., 2025). The results shown by Bergfeld et al., 2025 show that anticrack propagation dominates the early stage of crack propagation, but after a certain propagation distance, shear-driven fractures occur, especially on steeper slopes. This means that crack propagation leading to avalanches is a combination of both anticrack and shear fractures (Bergfeld et al., 2025). Furthermore, in supershear crack propagation in the simulations by Bergfeld et al., 2025, crack propagation was not arrested by slab fractures. This means that the release areas are much larger, because there is more extensive fracturing and faster propagation (Bergfeld et al., 2025). For sub-Rayleigh crack propagation, Bergfeld et al., 2025 observed crack arrest if there are existing slab fractures, which leads to smaller release areas. This is critical for the thesis because if crack propagation is slower, this means that the avalanche released is likely smaller. This is only a valid conclusion if the individual cracks can be observed

4.2 Numerical investigation of crack propagation regimes in snow fracture experiments (Bobillier et al., 2024)

Main Question of article: How do different snow mechanical parameters influence crack propagation speed?
Why this is relevant for the thesis: Understanding the influence between crack propagation speed and snow parameters is important to understand when cracks can be observed in DAS data if at all and how long it takes between initial cracking and avalanche release.
In previous models of crack propagation in snow, the mechanical properties of snow were not considered (Bobillier et al., 2024). A 3D discrete element model was developed by Bobillier et al., 2024 to investigate the mechanical properties of snow of crack propagation in a PST, this was done at different slope angles. Bobillier et al., 2024 assumed in their model that sintering and other time-dependent effects have no influence on crack propagation. A three-layered model consisting of a rigid bottom layer, a weak layer resembling depth hoar, as well as a dense slab layer was implemented (Bobillier et al., 2024). This type of model can possibly also be considered in the thesis. Understanding how different mechanical parameters affect crack propagation is relevant for the thesis because it influences if cracks can be observed in the DAS data at all, and if they are, how long it takes between initial cracking and slab breakoff, which is dependent on crack propagation speed. During the propagation saw test, the crack propagation speed

reaches steady state, the numerical simulation results by Bobillier et al., 2024 show that this is largely dependent on slope angle. At slope angles lower than 30°, the steady-state propagation speed is sub-Rayleigh, if the slope angle is greater or equal to 30°, the steady-state speed is larger than the shear wave speed, so supershear (Bobillier et al., 2024). It also depends on the slope angle how long it takes to reach this steady-state speed (Bobillier et al., 2024). Bobillier et al., 2024 show that the higher the slab elastic modulus, the lower the normalized crack propagation speed, this is the opposite for weak layer elastic modulus: the higher the weak layer elastic modulus, the higher the crack propagation speed. If the weak layer is very weak, crack propagation speed is also very low, Bobillier et al., 2024 explain this by failure occurring before cracks have propagated. Direct failure also occurs when the slab is very soft (Bobillier et al., 2024). These considerations are important for the thesis because it can mean that cracking might not be observed before the slab break off, or if the system is too stable, cracking is observed but there is no avalanche.

4.3 Transition from sub-Rayleigh anticrack to supershear crack propagation in snow avalanches (Trottet et al., 2022)

Main Question of article: What are the factors which influence cracks to propagate at speed faster than shear wave speed in snow slabs and how can this be modeled?
Why this is relevant for the thesis: Understanding at which speeds cracks propagate under different conditions and whether the cracks cause avalanches is important to make accurate risk assessments based on DAS data.
Previous to Trottet et al., 2022, two main theories about the propagation of cracks in snow slabs have been discussed. Some authors investigate shear only modes of failure, while others investigate anticrack modes (Trottet et al., 2022). In nature, large crack speeds higher than the velocity of shear waves were observed, while in smaller-scale experiments, sub-Rayleigh propagation was observed (Trottet et al., 2022). Understanding at which speeds cracks propagate at which modeling scales is important to accurately assess the risk associated with crack observations in DAS data: if cracks propagate at high speeds, slab breakoff will occur faster than if cracks propagate at lower speeds. Trottet et al., 2022 conducted multi point material modeling of propagation saw tests at different angles. Trottet et al., 2022 observed that the onset of supershear fractures depends mainly on terrain angle, on flatter terrain, the stress induced by slab bending causes propagation upslope, whereas in steep terrain, propagation was observed downslope because the normal stress is higher than shear stress. In inclinations which are lower than the effective friction of the material, the propagation speed was sub-Rayleigh, whereas at angles higher than the effective friction, supershear propagation was observed (Trottet et al., 2022). This was explained by Trottet et al., 2022 by slab bending. In flatter terrain, the bending of the slab is limited by the collapse depth (the slab can only bend as much as the snow layer collapses because the terrain is flat), which leads to a constant speed of crack propagation (Trottet et al., 2022). In inclined terrains, slab tension can increase, which also leads to crack propagation speeds to increase (Trottet et al., 2022). This is also an important consideration for DAS observations to assess risk correctly because of the time passed between initial crack and the breakoff of the slab. The slab considered in the simulations by Trottet et al., 2022 is pure elastic, at lower tensile strength of the snow this leads to initial cracking at the surface, crack arrest because of en echelon cracks and consequently smaller release areas. At higher tensile strengths in slabs, fractures occur at the bottom of the slab, the cracking does not influence crack propagation as much and release areas are larger (Trottet et al., 2022). Trottet et al., 2022 come to the conclusion that large slab avalanches with hard slabs (high tensile strength) show supershear cracks, they also mention that small-scale crack experiments are not representative of actual speeds.

4.4 Modeling snow slab avalanches caused by weak-layer failure - Part 2: Coupled mixed-mode criterion for skier-triggered anticracks (Rosendahl & Weißgraeber, 2020)

Main Question of article: How can anticracks in snow be described by different stress conditions (skier triggered) and how does critical stress needed for slab breakoff change with slope and snow parameters?
Why this is relevant for the thesis: Understanding what causes critical cracks to form is understand to predict where they can be observed in DAS data
Rosendahl and Weißgraeber, 2020 describe that classical models only account for growth of existing cracks, caused by instabilities in weak layers. The new model developed by Rosendahl and Weißgraeber, 2020 does not assume existing weak layers, here, a slab release only occurs if there is enough energy

for the crack to propagate and the critical stress is reached. These considerations are important for DAS data observations, because the snow stratigraphy might not be known in the area, which would mean that assumptions about existing weak layers cannot be made. Rosendahl and Weißgraeber, 2020 distinguish between two failure modes: Mode I, where cracks open normal to crack faces, and mode II where cracks open tangential to existing crack faces, the model developed accounts for both failure modes by considering the ratio between the shear and compressive force. Independent of the shear and compressive force ratio, Rosendahl and Weißgraeber, 2020 observe that the critical skier force needed for failure decreases with increasing slope angle. This means that in very steep terrain, an avalanche can also release without outside forces (Rosendahl & Weißgraeber, 2020). The numerical modeling results show that as the slab thickness increases, the critical skier force also increases, Rosendahl and Weißgraeber, 2020 explain this by the load distribution. Thicker slabs can distribute skier load more evenly, this is only true up until a certain thickness however, where the slab becomes too heavy (Rosendahl & Weißgraeber, 2020). The same is true for Young's modulus: the higher the Young's modulus of the slab, the stiffer and more resistant to deformation by the skier it is (Rosendahl & Weißgraeber, 2020). Rosendahl and Weißgraeber, 2020 explain this by lower Young's moduli slabs deforming more easily, which leads to more bending of the slab, which results in the load being concentrated on one part of the weak layer, which leads to easier failure. The new model developed by Rosendahl and Weißgraeber, 2020 observes that failure mode II dominates in more steep terrain, this observation is important when considering DAS data, because there might be directional effects between the orientation of the cracks and the cable in the data. The major limitation of the model, as stated by Rosendahl and Weißgraeber, 2020 is that the model relies on linear elastic solutions. This means that time-dependent processes, such as sintering are not taken into account

4.5 Investigating the release and flow of snow avalanches at the slope-scale using a unified model based on the material point method (Gaume et al., 2019)

Main Question of article: How can crack propagation and slab breakoff be modeled using the material point method in 2D and 3D?
Why this is relevant for the thesis: The material point method will also be used in the thesis, understanding how to model slab breakoff with this and which considerations need to be taken is important. Furthermore, it is important to understand how cracks propagate and how this can be modeled to understand and model DAS data of avalanches.
Gaume et al., 2019 state that snow can behave like a solid or like a fluid depending on the type of deformation which is present, modeling this behavior is important to understand which types of loading correspond to which behavior and deformation. The release of avalanches is because of failure initiation in a weak layer, the onset of crack propagation in the weak layer and the resulting detachment of the sliding slab (Gaume et al., 2019). Because the weak layer has porous character, there is volumetric collapse, which leads to the closure of cracks in the weak layer, so called anticrack behavior (Gaume et al., 2019). The new approach to model avalanche dynamics described by Gaume et al., 2019 uses the multi-point method to model deformation. Here, particles in the snowpack are used to track mass, momentum and deformation of the snowpack (Gaume et al., 2019). This means that individual particles instead of fixed mesh points are used for modeling, the lack of connectivity between the points complicates the calculation of spatial derivatives, so an Eulerian background is combined with interpolation functions by Gaume et al., 2019 to circumvent this problem. This is also an important consideration in case the MPM method will be used in the thesis. The model described by Gaume et al., 2019 uses a mixed-mode shear and compression yield surface and accounts for softening and hardening of the weak layer. The model accounts for the fact that the internal friction of snow is usually lower than the dynamic friction in the cracks, these are described by different parameters in the model (Gaume et al., 2019). The simulations run by Gaume et al., 2019 include slope-scale simulations in 2D and 3D with a constant slab depth, which account for spatial variability of the snow depth. The failure in the weak layer is initiated by loading at the bottom of the slope (Gaume et al., 2019). The failure propagates along the weak layer in a mixed-mode anticrack, the propagation speed modeled by Gaume et al., 2019 shows large variations. Changes in slope angle and curvature for example have large influences in the speed: concave leads to high propagation speed, whereas convex parts of the slope have lower speeds (Gaume et al., 2019). The propagation speed of the failure cracks is also influenced by slab tensile fractures: the closer to a slab tensile fracture, the lower the propagation speed (Gaume et al., 2019). The propagation speed of the failure cracks observed by Gaume et al., 2019 also showed strong directionality: propagation in the slope

direction (uphill) was observed to be 2x higher than in the downslope direction. Understanding the variations in propagation speed is very important to make sense of DAS observations, for example the time passed between loading and slab breakoff and how this can be explained. The slab is released where the slope angle exceeds the friction angle of the material (Gaume et al., 2019). This is an important consideration when thinking about the observation of slab breakoff in DAS data, because it can be assessed where the breakoff will happen: does it happen far away from the cable when signal could be attenuated or not? Gaume et al., 2019 state that the initial crown fracture of the slab breakoff is followed by a staunchwall and flank shear fractures. Could all of these different fractures be seen in DAS or are they maybe recorded as one event? Broken slab pieces can stick together again, the collective behavior of these particles leads to fluid-like behavior (Gaume et al., 2019). For remote triggering of avalanches, there needs to be the collapse of the weak layer as well as fracture propagation (Gaume et al., 2019). The collapse of the weak layer is negligible on steep slopes according to Gaume et al., 2019, the large angle is sufficient to observe failure by shearing alone. Here, weak layer collapse is secondary to failure Gaume et al., 2019. Could these separate processes also be observed? Like weak layer collapse would lead to deformation of DAS cable, which could be an indicator even before slab breakoff is observed, but of course only for flatter slopes. Gaume et al., 2019 observed a top to bottom collapse (from top of hill to bottom) in their simulations, which is consistent with large-scale PST experiments. The limitations of the model described by Gaume et al., 2019 is that the strength of the weak layer is not dependent on strain rate. This means that there is no potential for sintering as described by Mulak and Gaume, 2019, the model by Gaume et al., 2019 can therefore only be applied for fast loading, for example for failure induced by a skier. Another limitation of the model is that there is only variation in snow depth, other variations in other parameters are not accounted for by the Gaume et al., 2019 model.

4.6 Dynamic anticrack propagation in snow (Gaume et al., 2018)

Main Question of article: How can anticrack propagation in snow be modeled in a way which also represents strain softening accurately?
Why this is relevant for the thesis: Understanding how strain softening and crack propagation is modeled is important to make accurate models of DAS data
Models previous to Gaume et al., 2018 accounted for anticrack propagation by removing mesh elements. Gaume et al., 2018 developed a new model which accounts for the closure of cracks during anticrack propagation. Understanding this type of modeling is important for the thesis because anticracks are believed to be the main cause of snow slab avalanches. Modeling the correct type of material behavior and crack propagation is important to model DAS data accurately. Gaume et al., 2018 modeled hardening and softening of the slab by a yield surface with varying pressure, this leads the model to include strain-softening even at macroscopic uniaxial compression. Understanding snow material and model response is important to make sense of DAS data: which stages of deformation or material response can be identified? The results of the simulations show stages in material response: the first stage is an elastic regime, followed by a pressure and shear stress drop, then a volumetric collapse indicated by plastic consolidation, with the last stage being dense packing (Gaume et al., 2018). The results show important distinctions of slope simulations and propagation saw tests: Gaume et al., 2018 show that for slope simulations, failure occurs bottom to top (like remote triggering), whereas for smaller scale PST simulations, failure occurs top to bottom. This is a crucial consideration for simulation of DAS data because it shows where failure can be observed in terms of cable location. Gaume et al., 2018 explain the different propagation behavior by the strain caused by slab bending: at slope scales, the tensile stress caused by slab bending is not large enough to cause fracture, whereas in PST it is. Again, a MPM was used in combination with an Eulerian mesh to make time-dependent calculations easier.

4.7 Snow fracture in relation to slab avalanche release: critical state for the onset of crack propagation (Gaume et al., 2017)

Main Question of article: How can the critical crack length needed for avalanche release be modeled and which parameters influence this?
Why this is relevant for the thesis: The critical crack length and the parameters which influences this are important because it influences what can be seen in DAS data. This is in relation where data is measured: slope angle etc has an influence on how many avalanches are recorded by DAS and subsequently how large the dataset for breakoff observation is

Gaume et al., 2017 develop an equation which describes the relation of critical crack length to snowpack parameters. The release of a dry-snow slab avalanche requires the formation of a weak layer, the initial failure is a macroscopic crack in the weak layer (Gaume et al., 2017). The length of this crack cannot exceed the critical crack length, because else there will be material failure and avalanche release (Gaume et al., 2017). This is because the critical crack length is the length of the crack which is present when the shear stress is equal to the shear strength of the material. Understanding the influences of different snow and terrain parameters on critical crack length is relevant for DAS observations of avalanche breakoff because it can be understood how many avalanches occur and why. Previous modeling of avalanche release to Gaume et al., 2017 made simplifications to crack propagation, which results in inaccurate predictions. Gaume et al., 2017 estimate critical crack length using a discrete element model as described by Cundall and Strack, 1979, where the cracking is due to sawing in the weak layer, similar to Gaume et al., 2015. The modeled data is then compared to experimental data. The numerical simulations by Gaume et al., 2017 show that critical crack length increases, if elastic modulus of the slab increases or the weak layer thickness increases. The critical crack length on the other hand decreases if the slab density increases, the slab thickness increases or the slope angle increases (Gaume et al., 2017), this means that under these conditions, cracks need to be less long for failure to occur. Gaume et al., 2017 state that the shear stress at the crack tip can be decomposed into a part caused by slab tension and a second part caused by slab bending. The analytical expression for critical crack length taking into account snow and terrain parameters agree with experimental results of saw tests conducted by Gaume et al., 2017. The length of the critical crack is also relevant for the thesis because maybe shorter cracks show up less well in DAS data (less of a displacement). The analytical expression derived describes critical crack length better than the anticrack model, especially in regard to weak layer thickness and slope angle (Gaume et al., 2017). The anticrack model shows that critical crack length is not dependent on slope angle, this is because the model makes assumptions for simplification which the new expression by Gaume et al., 2017 does not make. The new expression derived by Gaume et al., 2017 has some limitations, the dependence of critical length on slope angle could also be caused by young's modulus and slab thickness, if these two are decreased, the critical length decreases less with increasing slope angle. Furthermore, Gaume et al., 2017 do not account for the shear stresses caused by slab bending, but suggest that this effect is small because simulation results are in good agreement with observations. The models described here assume a homogenous slab, where in reality the snowpack has multiple layers of different strengths, which can have an effect (Gaume et al., 2017).

4.8 Modeling of crack propagation in weak snowpack layers using the discrete element method (Gaume et al., 2015)

Main Question of article: Which snow parameters influence crack propagation distance and speed and how can these processes be modeled?
Why this is relevant for the thesis: Understanding the influence of different snow parameters on crack propagation distance and speed is important to model and understand slab breakoff DAS data in thesis.
Gaume et al., 2015 state that dry-snow slab avalanches form by the failure of a weak snow layer underlying a cohesive slab. The local damage leading to failure is a crack in the weak layer, which when it reaches critical length leads to tensile fracture in the slab, which leads to avalanche release (Gaume et al., 2015). Understanding how these processes can be modeled and which parameters affect the propagation distance and speed is important for the thesis because slab breakoff is investigated in DAS data. Understanding which snow parameters lead to large crack propagation or speed is important to determine in which DAS datasets slab breakoff is most likely to be detected and at which times in the dataset this would be (does the breakoff take a lot of time or not because speeds and distances are low or high?). Gaume et al., 2015 state that the dynamic phase of crack propagation and the effects of different snow parameters are poorly understood, which limits avalanche forecasting because it is unclear how and under which circumstances cracks propagate. This is directly linked to the thesis, because cracking of the snow layer is investigated in DAS data. The method used by Gaume et al., 2015 to model cracking in snow is the discrete element method described by Cundall and Strack, 1979, with a limitation that the slab is modeled as a continuous material with elastic-brittle behavior. The loading modeled is due to gravity and the propagation saw test (Gaume et al., 2015), the cohesive part of the snow (so the connections between the grains) are modeled by the linear contact law, the maximum and minimum and minimum tensile and shear stresses are modeled by beam theory (Gaume et al., 2015). The stiffness at the bonded contact (of grains) is calculated (Gaume et al., 2015), this is important to see how the material reacts: what influence does

the stiffness of the grain contacts have on crack propagation distance and speed? Gaume et al., 2015 calculate the time-step of the simulations from the snow grain properties, this ensures that the maximum crack propagation velocity is lower than the speed of elastic waves in the snow, this means that the cracks in the snow cannot propagate faster than the propagation of the elastic waves which are caused by the loading. Understanding the relationship between elastic wave velocity and crack propagation speed is also important for the modeling of DAS data of slab breakoff which will be done in the thesis. Gaume et al., 2015 determined the propagation distance of cracks in the slab by setting the tensile and strength to finite values, to determine propagation speed on the other hand, these values were set to infinity. Understanding the propagation speed and distance of cracks is important because the higher either of these values, the faster the avalanche breaks off. The propagation distance described by Gaume et al., 2015 is defined as the distance between the point where failure initiates, so the point where the initial load is applied and the point where the tensile failure is located. The numerical models showed that the crack propagation speed increases with increasing young's modulus, increasing thickness of slab and increasing density, but the thickness of weak layer has no influence on propagation speed (Gaume et al., 2015). From their results, Gaume et al., 2015 suggest that the lower the young's modulus of a snow layer, meaning the softer a snow layer, the lower the crack propagation velocity, which means that avalanches break off less fast. This suggests that the propagation speed of cracks is mainly influenced by failure conditions such as the tensile strength of the snow layer rather than structural parameters such as weak layer thickness (Gaume et al., 2015), which is relevant for the thesis because factors such as weak layer thickness need not be modeled, which saves computational time. The propagation distance of cracks is influenced by weak layer thickness however, Gaume et al., 2015 suggest that if weak layer thickness increases, there is more slab bending which leads to a higher tensile stress and more likely tensile failure, which means that crack propagation distance decreases, because failure happens before the cracks can propagate very far (Gaume et al., 2015). Furthermore, the simulations run by Gaume et al., 2015 show that the slope angle has little influence on crack propagation for low tensile strength snow layers, but for high tensile strength layers there is a large correlation between slope angle and propagation distance of cracks (Gaume et al., 2015). This can be explained by the main failure mode being tensile failure due to slab bending at high tensile stresses of the snow layer (Gaume et al., 2015). Gaume et al., 2015 conducted lab experiments, which confirm the results of the numerical simulations in all but one case: the relationship between high young's modulus and large crack propagation is counterintuitive, because in harder snow, cracks propagate less well compared to soft snow, Gaume et al., 2015 explain this by the limitations of the beam theory to account for dynamic effects. This is an important consideration depending on which method will be used to model crack propagation in the DAS data. Gaume et al., 2015 show that the stress evolution in the snowpack always shows an increase of tensile stress at the top of the slab before crack propagation, which happens close to the tip of the crack in the weak layer. The tensile crack opens up as soon as the tensile stress is equal to the tensile strength of the snow, this crack always starts at the top surface of the slab (Gaume et al., 2015). Understanding the stress evolution of the snowpack is important for DAS data interpretation, because it is possible that the different phases of this path (increase in stress, opening of crack in weak layer, tensile cracking and failure) can be seen in the data. Gaume et al., 2015 mention two important limitations of the simulation results: the models assume a simplified shape of the weak layer, as well as the assumption that the crack propagation speed is lower than the speed of the elastic wave. At high propagation speeds of the crack, the tangential displacement of the snowpack becomes more important than the vertical displacement (Gaume et al., 2015). Furthermore, the column length of the modeled propagation saw test, which results in loading of the modeled snow, has large influences on the simulation results depending on the snow properties (Gaume et al., 2015), the thicker and stronger the snow layer, the larger the column must be to get accurate results. To summarize the snow parameters important for propagation distance and speed, which are important for simulations of slab breakoff in the thesis:

- propagation speed increases if: young's modulus increases, slab depth increases, slab density increases or the slope angle increases
- propagation velocity decreases if weak layer has increasing strength
- propagation distance decreases if young's modulus, slab density or weak layer thickness increases (leads to faster failure)
- propagation distance increases if slab tensile strength, slab depth, weak layer strength or slope angle increases

- For very dense slab layers, tensile fracture and avalanche release is affected by topography (trees, large rocks in the avalanche path, or heterogeneity of snowcover)

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